

Лабораторная работа №7

Адресация IPv4 и IPv6. Настройка DHCP

Ромицына А. Р.

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Российский университет дружбы народов, Москва, Россия

Информация

- Ромицына Анастасия Романовна
- НПИбд-02-23 Студ. билет: 1132236132
- Российский университет дружбы народов
- 1132236132@pfur.ru

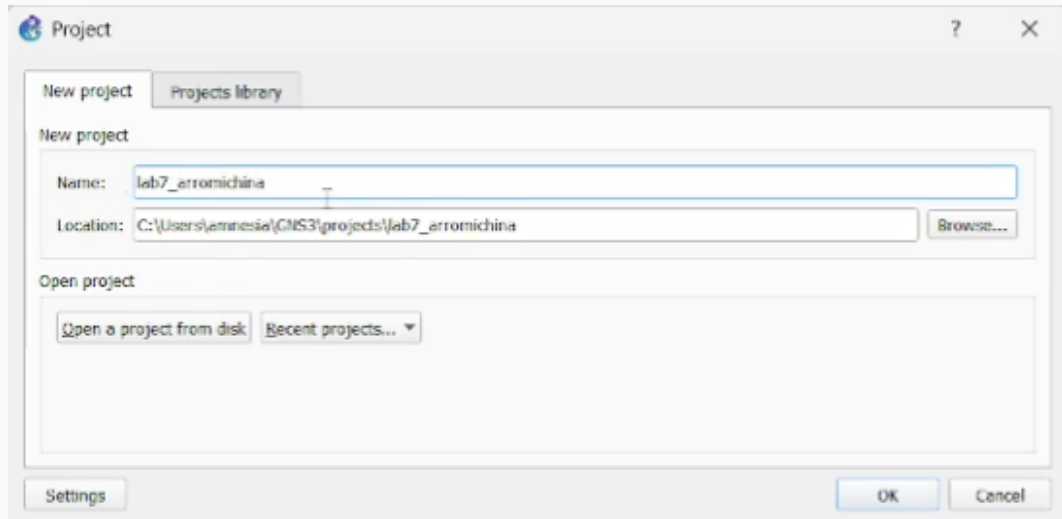
Вводная часть

- Целью данной работы является получение навыков настройки службы DHCP на сетевом оборудовании для распределения адресов IPv4 и IPv6.

Основная часть

Настройка DHCP в случае IPv4

- Создание нового проекта в GNS3.



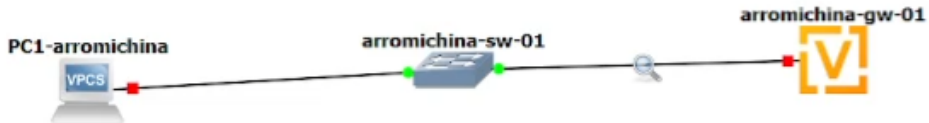
Настройка DHCP в случае IPv4

- Размещение устройств в соответствии с топологией в лабораторной работе. Изменение названий устройств.



Настройка DHCP в случае IPv4

- Включение захвата трафика.



Настройка DHCP в случае IPv4

- Настройка образа VyOS и переход на маршрутизаторе в режим конфигурирования, изменение имени устройства и доменного имени, замена системного пользователя.

```
vyos@vyos:~$ configure
[edit]
vyos@vyos# set system host-name arromichina-gw-01
[edit]
vyos@vyos# set system domain-name arromichina.net
[edit]
vyos@vyos# set system login user arromichina
[edit]
[edit]
123456yos# set system login user arromichina authentication plaintext-password
[edit]
vyos@vyos# commit
[edit]
vyos@vyos# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@vyos# exit
exit
vyos@vyos:~$ exit
logout
```

Welcome to VyOS - arromichina-gw-01 ttyS0

arromichina-gw-01 login: arromichina

Password:

Welcome to VyOS!

Check out project news at <https://blog.vyos.io>
and feel free to report bugs at <https://vyos.dev>

You can change this banner using "set system login banner post-login" command.

VyOS is a free software distribution that includes multiple components.

Настройка DHCP в случае IPv4

- Настройка адресации IPv4, добавление конфигурации DHCP-сервера на маршрутизаторе.

```
arromichina@arromichina-gw-01# set interfaces ethernet eth0 address 10.0.0.1/24
[edit]
arromichina@arromichina-gw-01# commit
[edit]
arromichina@arromichina-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
ichina domain-name arromichina.net service dhcp-server shared-network-name arrom
[edit]
ichina name-server 10.0.0.101# set service dhcp-server shared-network-name arrom
[edit]
ichina subnet 10.0.0.0/24w-01# set service dhcp-server shared-network-name arrom
[edit]
ichina subnet 10.0.0.0/24 default-router 10.0.0.1erver shared-network-name arromi
[edit]
ichina subnet 10.0.0.0/24 range hosts start 10.0.0.2er shared-network-name arrom
[edit]
ichina subnet 10.0.0.0/24 range hosts stop 10.0.0.253r shared-network-name arrom
[edit]
arromichina@arromichina-gw-01# commit
[edit]
arromichina@arromichina-gw-01# save
```

Настройка DHCP в случае IPv4

- Просмотр статистики DHCP-сервера и выданных адресов.

```
arromichina@arromichina-gw-01:~$ show dhcp server statistics
Pool          Size      Leases    Available  Usage
-----
arromichina    252        0         252       0%
arromichina@arromichina-gw-01:~$ show dhcp server leases
IP address    Hardware address  State    Lease start    Lease expiration  Remaining  P
ool           Hostname
-----
-----
arromichina@arromichina-gw-01:~$
```

Настройка DHCP в случае IPv4

- Настройка оконечного устройства PC1-arromichina.

```
PC1-arromichina - PuTTY
68:00

Opcode: 2 (REPLY)
Client IP Address: 0.0.0.0
Your IP Address: 10.0.0.2
Server IP Address: 0.0.0.0
Gateway IP Address: 0.0.0.0
Client MAC Address: 00:50:79:66:68:00
Option 53: Message Type = Offer
Option 54: DHCP Server = 10.0.0.1
Option 51: Lease Time = 86400
Option 1: Subnet Mask = 255.255.255.0
Option 3: Router = 10.0.0.1
Option 6: DNS Server = 10.0.0.1
Option 15: Domain = arromichina.net

Opcode: 1 (REQUEST)
Client IP Address: 10.0.0.2
Your IP Address: 0.0.0.0
Server IP Address: 0.0.0.0
Gateway IP Address: 0.0.0.0
Client MAC Address: 00:50:79:66:68:00
Option 53: Message Type = Request
Option 54: DHCP Server = 10.0.0.1
Option 50: Requested IP Address = 10.0.0.2
Option 61: Client Identifier = Hardware Type=Ethernet MAC Address = 00:50:79:66:
68:00
Option 12: Host Name = VPCS1

Opcode: 2 (REPLY)
Client IP Address: 10.0.0.2
Your IP Address: 10.0.0.2
Server IP Address: 0.0.0.0
Gateway IP Address: 0.0.0.0
Client MAC Address: 00:50:79:66:68:00
Option 53: Message Type = Ack
Option 54: DHCP Server = 10.0.0.1
Option 51: Lease Time = 86400
Option 1: Subnet Mask = 255.255.255.0
Option 3: Router = 10.0.0.1
Option 6: DNS Server = 10.0.0.1
Option 15: Domain = arromichina.net

IP 10.0.0.2/24 GW 10.0.0.1
```

Настройка DHCP в случае IPv4

- Проверка конфигурации IPv4 на узле и пинг маршрутизатора.

```
VPCS> show ip

NAME           : VPCS[1]
IP/MASK        : 10.0.0.2/24
GATEWAY        : 10.0.0.1
DNS            : 10.0.0.1
DHCP SERVER    : 10.0.0.1
DHCP LEASE     : 86362, 86400/43200/75600
DOMAIN NAME    : arromichina.net
MAC            : 00:50:79:66:68:00
LPORT          : 10003
RHOST:PORT     : 127.0.0.1:10004
MTU:           : 1500

VPCS> ping 10.0.0.1 -c 2
64 bytes from 10.0.0.1 icmp_seq=1 ttl=64 time=0.756 ms
64 bytes from 10.0.0.1 icmp_seq=2 ttl=64 time=2.301 ms

VPCS> 
```

Настройка DHCP в случае IPv4

- Просмотр статистики DHCP-сервера и выданных адресов.

```
arromichina@arromichina-gw-01:~$ show dhcp server statistics
Pool          Size      Leases    Available  Usage
-----
arromichina    252        1         251       0%
arromichina@arromichina-gw-01:~$ show dhcp server leases
IP address    Hardware address  State    Lease start          Lease expiration      Remain
ng   Pool      Hostname
-----
10.0.0.2      00:50:79:66:68:00  active   2025/12/05 19:55:36  2025/12/06 19:55:36  23:58:
5    arromichina  VPCS1
arromichina@arromichina-gw-01:~$ █
```

Настройка DHCP в случае IPv4

- Просмотр журнала работы DHCP-сервера.

```
arromichina@arromichina-gw-01:~$ show log | grep dhcp
Dec 05 19:39:58 dhclient-script-vyos[1966]: Deleting search domains with tag "dhcp-eth0" via
vyos-hostsd-client
Dec 05 19:39:59 vyos-hostsd[704]: Request data: {"type": "search_domains", "op": "delete", "
data": ["dhcp-eth0"]}
Dec 05 19:39:59 dhclient-script-vyos[1966]: Deleting nameservers with tag "dhcp-eth0" via vy
os-hostsd-client
Dec 05 19:40:00 vyos-hostsd[704]: Request data: {"type": "name_servers", "op": "delete", "da
ta": ["dhcp-eth0"]}
Dec 05 19:41:08 dhclient-script-vyos[2508]: Deleting search domains with tag "dhcp-eth0" via
vyos-hostsd-client
Dec 05 19:41:09 vyos-hostsd[704]: Request data: {"type": "search_domains", "op": "delete", "
data": ["dhcp-eth0"]}
Dec 05 19:41:09 dhclient-script-vyos[2508]: Deleting nameservers with tag "dhcp-eth0" via vy
os-hostsd-client
Dec 05 19:41:09 vyos-hostsd[704]: Request data: {"type": "name_servers", "op": "delete", "da
ta": ["dhcp-eth0"]}
Dec 05 19:43:29 dhclient-script-vyos[2693]: Deleting search domains with tag "dhcp-eth0" via
vyos-hostsd-client
Dec 05 19:43:30 vyos-hostsd[704]: Request data: {"type": "search_domains", "op": "delete", "
data": ["dhcp-eth0"]}
Dec 05 19:43:30 dhclient-script-vyos[2693]: Deleting nameservers with tag "dhcp-eth0" via vy
os-hostsd-client
Dec 05 19:43:30 vyos-hostsd[704]: Request data: {"type": "name_servers", "op": "delete", "da
ta": ["dhcp-eth0"]}
Dec 05 19:45:06 dhclient-script-vyos[2677]: Deleting search domains with tag "dhcp-eth0" via
```


Настройка DHCP в случае IPv4

- Проанализируем захваченные анализатором трафика пакеты

The screenshot shows the Wireshark interface with a packet list on the left and a packet details pane on the right. The packet list shows a series of DHCP Discover packets from 0.0.0.0 to 255.255.255.255. The selected packet is number 11, a DHCP Discover packet from 0.0.0.0 to 255.255.255.255.

Time	Source	Destination	Protocol	Length	Info
1 0.000000	::::	ff02::2	ICMPv6	62	Router Solicitation
2 156.143200	::	ff02::16	ICMPv6	130	Multicast Listener Report Message v2
3 156.782069	::	ff02::1:ff72:0	ICMPv6	86	Neighbor Solicitation for fe80::eae:92ff:fe72:0
4 157.160738	::	ff02::16	ICMPv6	130	Multicast Listener Report Message v2
5 157.812076	fe80::eae:92ff:fe72...	ff02::16	ICMPv6	150	Multicast Listener Report Message v2
6 157.815719	fe80::eae:92ff:fe72...	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
7 158.047257	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x10fc7c0e
8 158.220299	fe80::eae:92ff:fe72...	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
9 158.693380	fe80::eae:92ff:fe72...	ff02::16	ICMPv6	150	Multicast Listener Report Message v2
10 169.061101	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0xe07f0932
11 172.132630	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0xe07f0932
12 174.195179	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0xe07f0932
13 179.451752	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0xe07f0932
14 189.945506	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0xe07f0932
15 201.399014	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0xe07f0932
16 215.821006	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0xe07f0932
17 226.136874	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0xe07f0932
18 309.207134	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0xbea7ca2c
19 312.199125	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0xbea7ca2c
20 318.683083	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0xbea7ca2c
21 332.651870	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0xbea7ca2c
22 343.641986	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0xbea7ca2c
23 357.689732	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0xbea7ca2c
24 369.937731	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0xbea7ca2c
25 426.326312	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0xe0d284e
26 429.229041	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0xe0d284e

Packet 11 details:

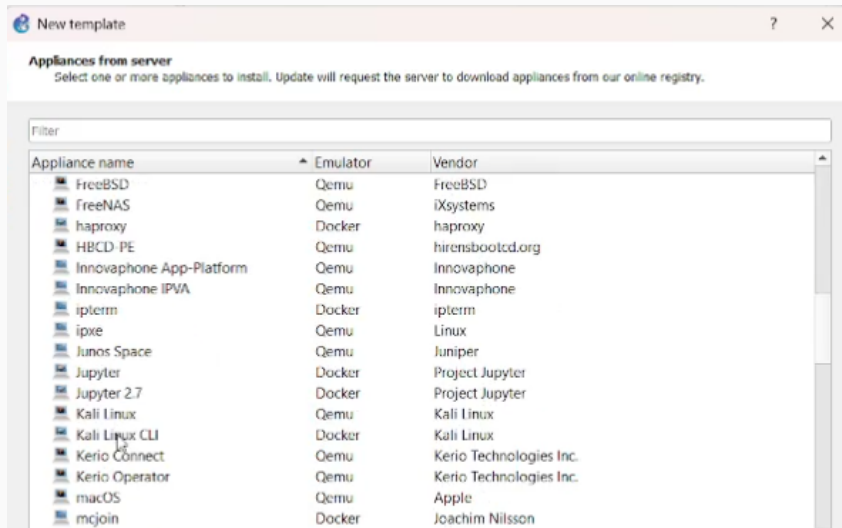
```
Frame 11: 342 bytes on wire (2736 bits), 342 bytes captured (2736 bits) on interface -, id 0
Ethernet II, Src: 0c:a6:92:72:00:00 (0c:a6:92:72:00:00), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
Internet Protocol Version 4, Src: 0.0.0.0, Dst: 255.255.255.255
User Datagram Protocol, Src Port: 68, Dst Port: 67
Dynamic Host Configuration Protocol (Discover)
```

Hex dump of packet 11:

```
0000 ff ff ff ff ff ff 0c ae 92 72 00 00 00 00 45 16
0010 01 48 00 00 00 00 00 11 39 96 00 00 00 00 ff ff
0020 ff ff 00 44 00 43 01 24 44 43 01 01 06 00 e0 7f
0030 09 32 00 03 00 00 00 00 00 00 00 00 00 00 00 00
0040 00 00 00 00 00 00 0c ae 92 72 00 00 00 00 00 00
0050 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0060 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0070 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0080 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0090 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00a0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00b0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00c0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00d0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
```

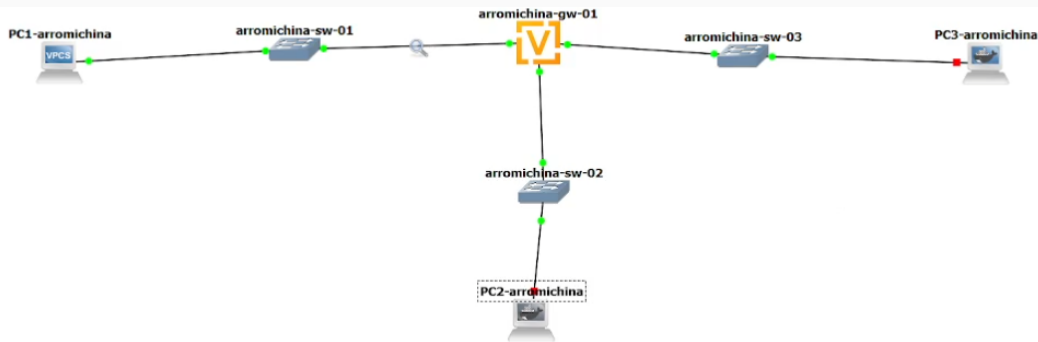
Настройка DHCP в случае IPv6

- Добавление образа Kali Linux CLI в перечень устройств в GNS3.



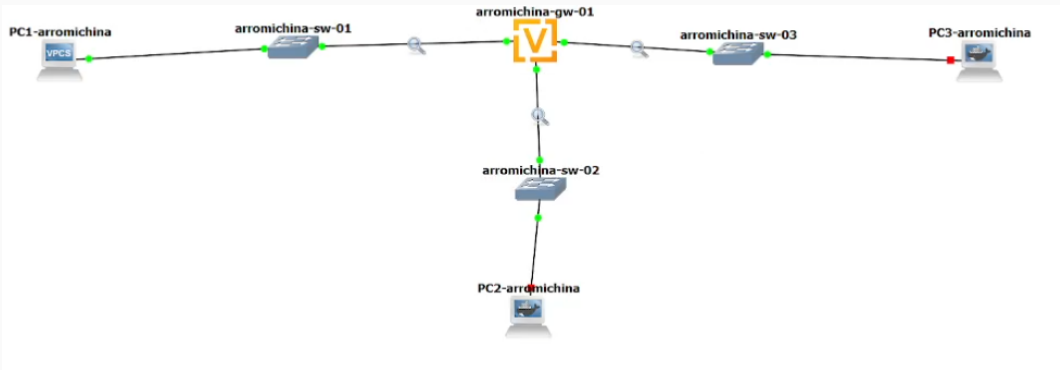
Настройка DHCP в случае IPv6

- Дополнение сети в соответствии с топологией, приведённой в лабораторной работе. Изменение отображаемых названий устройств.



Настройка DHCP в случае IPv6

- Включение захвата трафика.



Настройка DHCP в случае IPv6

- Настройка адресации IPv6 на маршрутизаторе.

```
arromichina-gw-01 - PuTTY
MMAND=/usr/libexec/vyos/op_mode/show_dhcp.py --leases
arromichina@arromichina-gw-01:~$ configure
[edit]
arromichina@arromichina-gw-01# set interfaces ethernet eth1 address 2000::1/64
[edit]
arromichina@arromichina-gw-01# set interfaces ethernet eth2 address 2001::1/64
[edit]
arromichina@arromichina-gw-01# show interfaces
 ethernet eth0 {
   address 10.0.0.1/24
   hw-id 0c:ae:92:72:00:00
 }
 ethernet eth1 {
+  address 2000::1/64
   hw-id 0c:ae:92:72:00:01
 }
 ethernet eth2 {
+  address 2001::1/64
   hw-id 0c:ae:92:72:00:02
 }
 ethernet eth3 {
   hw-id 0c:ae:92:72:00:03
 }
 ethernet eth4 {
   hw-id 0c:ae:92:72:00:04
 }
 ethernet eth5 {
   hw-id 0c:ae:92:72:00:05
 }
 ethernet eth6 {
   hw-id 0c:ae:92:72:00:06
 }
 ethernet eth7 {
   hw-id 0c:ae:92:72:00:07
 }
 ethernet eth8 {
   hw-id 0c:ae:92:72:00:08
 }
 ethernet eth9 {
   hw-id 0c:ae:92:72:00:09
 }
 loopback lo {
```

Настройка DHCP в случае IPv6

- Настройка на маршрутизаторе DHCPv6 без отслеживания состояния.
Добавление конфигурации DHCP-сервера.

```
arromichina-gw-01 - PuTTY
onfig-flage@arromichina-gw-01# set service router-advert interface eth1 other-c
[edit]
omichina-stateless
[edit]
omichina -stateless subnet 2000::0/64rvice dhcpv6-server shared-network-name arr

Configuration path: service dhcpv6-server shared-network-name arromichina [-stateless] is
not valid
Set failed

[edit]
omichina-stateless subnet 2000::0/64ervice dhcpv6-server shared-network-name ar
[edit]
omichina-stateless common-options name-server 2000::1ver shared-network-name arr
[edit]
omichina-stateless common-options domain-search arromichina.net-network-name arr
[edit]
arromichina@arromichina-gw-01# commit
[edit]
arromichina@arromichina-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
arromichina@arromichina-gw-01# run show configuration
interfaces {
    ethernet eth0 {
        address 10.0.0.1/24
        hw-id 0c:ae:92:72:00:00
    }
    ethernet eth1 {
        address 2000::1/64
        hw-id 0c:ae:92:72:00:01
    }
    ethernet eth2 {
        address 2001::1/64
        hw-id 0c:ae:92:72:00:02
    }
    ethernet eth3 {
```

Настройка DHCP в случае IPv6

- Проверка на узле PC2-arromichina настройки сети.

```
PC2-arromichina - PuTTY
eth1    Link encap:Ethernet  HWaddr 02:42:52:B6:01:01
        inet6 addr: fe80::42:52ff:feb6:101/64 Scope:Link
        UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
        RX packets:0 errors:0 dropped:0 overruns:0 frame:0
        TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
        collisions:0 txqueuelen:1000
        RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)

lo      Link encap:Local Loopback
        inet addr:127.0.0.1  Mask:255.0.0.0
        inet6 addr: ::1/128 Scope:Host
        UP LOOPBACK RUNNING  MTU:65536  Metric:1
        RX packets:0 errors:0 dropped:0 overruns:0 frame:0
        TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
        collisions:0 txqueuelen:1000
        RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)

/ # route -n -A inet6
Kernel IPv6 routing table
Destination                                Next Hop                                Flags
Metric Ref    Use Iface
2000::/64
256 1         0 eth0      ::
UA
fe80::/64
256 1         0 eth0      ::
U
fe80::/64
256 1         0 eth1      ::
UGDA
::/0
1024 1        0 eth0      fe80::eae:92ff:fe72:1
Un
::1/128
0 3          0 lo        ::
Un
2000::42:52ff:feb6:100/128
0 2          0 eth0      ::
Un
fe80::42:52ff:feb6:100/128
0 2          0 eth0      ::
Un
fe80::42:52ff:feb6:101/128
0 3          0 eth1      ::
U
ff00::/8
256 3          0 eth0      ::
```

Настройка DHCP в случае IPv6

- Пинг маршрутизатора на узле PC2-arromichina, проверка настройки DNS, получение адреса по DHCPv6.

```
/ # ping 2000::1 -c 2
PING 2000::1 (2000::1): 56 data bytes
64 bytes from 2000::1: seq=0 ttl=64 time=2.209 ms
64 bytes from 2000::1: seq=1 ttl=64 time=6.707 ms

--- 2000::1 ping statistics ---
2 packets transmitted, 2 packets received, 0% packet loss
round-trip min/avg/max = 2.209/4.458/6.707 ms
/ # cat /etc/resolv.conf
/ # dhclient -6 -S -v eth0
```


Настройка DHCP в случае IPv6

- Пинг маршрутизатора от узла PC2-arromichina и проверка настройки DNS.

```
/ # ping 2000::1 -c2
PING 2000::1 (2000::1): 56 data bytes
64 bytes from 2000::1: seq=0 ttl=64 time=1.435 ms
64 bytes from 2000::1: seq=1 ttl=64 time=1.073 ms

--- 2000::1 ping statistics ---
2 packets transmitted, 2 packets received, 0% packet loss
round-trip min/avg/max = 1.073/1.254/1.435 ms
/ # cat /etc/resolv.conf
```

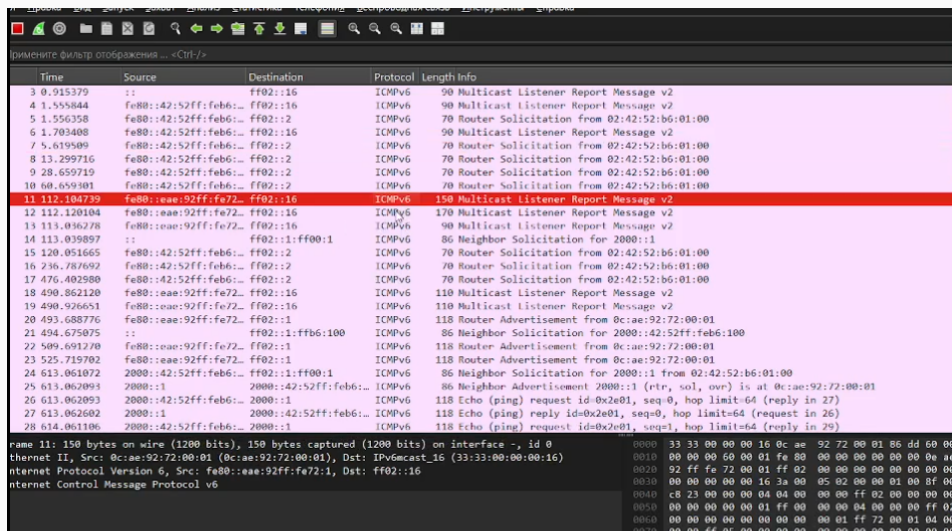
Настройка DHCP в случае IPv6

- Просмотр статистики на маршрутизаторе DHCP-сервера и выданных адресов.

```
arromichina@arromichina-gw-01# run show dhcpv6 server leases
IPv6 address      State      Last communication    Lease expiration      Remaining      Type      Pool
  IAID_DUID
-----
[edit]
arromichina@arromichina-gw-01#
```

Настройка DHCP в случае IPv6

- Проанализируем захваченные анализатором трафика пакеты



Примените фильтр отображения ... <Ctrl-F>

Time	Source	Destination	Protocol	Length	Info
3 0.915379	::	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
4 1.555844	fe80::42:52ff:feb6::...	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
5 1.556358	fe80::42:52ff:feb6::...	ff02::2	ICMPv6	70	Router Solicitation from 02:42:52:b6:01:00
6 1.703408	fe80::42:52ff:feb6::...	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
7 5.619509	fe80::42:52ff:feb6::...	ff02::2	ICMPv6	70	Router Solicitation from 02:42:52:b6:01:00
8 13.299716	fe80::42:52ff:feb6::...	ff02::2	ICMPv6	70	Router Solicitation from 02:42:52:b6:01:00
9 28.659719	fe80::42:52ff:feb6::...	ff02::2	ICMPv6	70	Router Solicitation from 02:42:52:b6:01:00
10 60.659301	fe80::42:52ff:feb6::...	ff02::2	ICMPv6	70	Router Solicitation from 02:42:52:b6:01:00
11 112.104739	fe80::eae:92ff:fe72::...	ff02::16	ICMPv6	150	Multicast Listener Report Message v2
12 112.120104	fe80::eae:92ff:fe72::...	ff02::16	ICMPv6	170	Multicast Listener Report Message v2
13 113.036278	fe80::eae:92ff:fe72::...	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
14 113.039897	::	ff02::1:ff00:1	ICMPv6	86	Neighbor Solicitation for 2000::1
15 120.051665	fe80::42:52ff:feb6::...	ff02::2	ICMPv6	70	Router Solicitation from 02:42:52:b6:01:00
16 236.787692	fe80::42:52ff:feb6::...	ff02::2	ICMPv6	70	Router Solicitation from 02:42:52:b6:01:00
17 476.402980	fe80::42:52ff:feb6::...	ff02::2	ICMPv6	70	Router Solicitation from 02:42:52:b6:01:00
18 490.862120	fe80::eae:92ff:fe72::...	ff02::16	ICMPv6	110	Multicast Listener Report Message v2
19 490.926651	fe80::eae:92ff:fe72::...	ff02::16	ICMPv6	110	Multicast Listener Report Message v2
20 493.688776	fe80::eae:92ff:fe72::...	ff02::1	ICMPv6	118	Router Advertisement from 0c:ae:92:72:00:01
21 494.675075	::	ff02::1:ffb6:100	ICMPv6	86	Neighbor Solicitation for 2000::42:52ff:feb6:100
22 509.691270	fe80::eae:92ff:fe72::...	ff02::1	ICMPv6	118	Router Advertisement from 0c:ae:92:72:00:01
23 525.719702	fe80::eae:92ff:fe72::...	ff02::1	ICMPv6	118	Router Advertisement from 0c:ae:92:72:00:01
24 613.061072	2000::42:52ff:feb6::...	ff02::1:ff00:1	ICMPv6	86	Neighbor Solicitation for 2000::1 from 02:42:52:b6:01:00
25 613.062093	2000::1	2000::42:52ff:feb6::...	ICMPv6	86	Neighbor Advertisement 2000::1 (rtr, sol, ovr) is at 0c:ae:92:72:00:01
26 613.062093	2000::42:52ff:feb6::...	2000::1	ICMPv6	118	Echo (ping) request id=0x2e01, seq=0, hop limit=64 (reply in 27)
27 613.062602	2000::1	2000::42:52ff:feb6::...	ICMPv6	118	Echo (ping) reply id=0x2e01, seq=0, hop limit=64 (request in 26)
28 614.061106	2000::42:52ff:feb6::...	2000::1	ICMPv6	118	Echo (ping) request id=0x2e01, seq=1, hop limit=64 (reply in 29)

name 11: 150 bytes on wire (1200 bits), 150 bytes captured (1200 bits) on interface -, id 0
ethernet II, Src: 0c:ae:92:72:00:01 (0c:ae:92:72:00:01), Dst: IPv6mcast_16 (33:33:00:00:00:16)
Internet Protocol Version 6, Src: fe80::eae:92ff:fe72:1, Dst: ff02::16
Internet Control Message Protocol v6

0000 33 33 00 00 00 16 0c ae 92 72 00 01 86 dd 50 00
0010 00 00 00 60 00 01 fe 80 00 00 00 00 00 00 0e a4
0020 92 ff fe 72 00 01 ff 02 00 00 00 00 00 00 00 00
0030 00 00 00 00 00 16 3a 00 05 02 00 00 01 00 8f 04
0040 c8 23 00 00 00 04 04 00 00 00 ff 02 00 00 00 00
0050 00 00 00 00 00 01 ff 00 00 00 04 00 00 ff 07
0060 00 00 00 00 00 00 00 00 00 00 01 ff 72 00 01 04 00
0070 00 00 66 06 00 00 00 00 00 02 00 02 00 00 00 00

Настройка DHCP в случае IPv6

- Настройка на маршрутизаторе DHCPv6 с отслеживанием состояния. Настройка на интерфейсе eth2 маршрутизатора объявления о маршрутизаторах. Добавление конфигурации DHCP-сервера на маршрутизатор.

```
flagmichina@arromichina-gw-01# set service router-advert interface eth2 managed-
[edit]
omichina-statefulichina-gw-01# set service dhcpv6-server shared-network-name arr
[edit]
omichina-stateful subnet 2001::0/64
[edit]
omichina-stateful subnet 2001::0/64 name-server 2001::1r shared-network-name arr
[edit]
omichina-stateful subnet 2001::0/64 domain-search arromichina.net
[edit]
[edit]na-stateful subnet 2001::0/64 address-range start 2001::100 stop 2001::199
arromichina@arromichina-gw-01# commit
[edit]
arromichina@arromichina-gw-01# save
Saving configuration to '/config/config.boot'...
Done
```

- Просмотр статистики DHCP-сервера на маршрутизаторе и выданных адресов.

```
arromichina@arromichina-gw-01# run show dhcpv6 server leases
IPv6 address      State      Last communication  Lease expiration  Remaining  Type  Pool
  IAID_DUID
-----
[edit]
arromichina@arromichina-gw-01#
```

Настройка DHCP в случае IPv6

- Подключение к узлу PC3-arromichina и проверка настройки сети.

```
BusyBox v1.30.1 (Ubuntu 1:1.30.1-4ubuntu6.5) built-in shell (ash)
Enter 'help' for a list of built-in commands.

/ # ifconfig
eth0      Link encap:Ethernet  HWaddr 02:42:44:54:2A:00
          inet6 addr: fe80::42:44ff:fe54:2a00/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:9 errors:0 dropped:0 overruns:0 frame:0
          TX packets:14 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:974 (974.0 B)  TX bytes:1076 (1.0 KiB)

eth1      Link encap:Ethernet  HWaddr 02:42:44:54:2A:01
          inet6 addr: fe80::42:44ff:fe54:2a01/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:65536  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)

/ # route -n -A inet6
Kernel IPv6 routing table

```

Destination				Next Hop	Fla
gs Metric Ref	Use	Iface			
fe80::/64				::	U
256 1	0	eth0			
fe80::/64				::	U
256 1	0	eth1			
::/0				fe80::eae:92ff:fe72:2	UGD
A 1024 1	0	eth0			
::1/128				::	Un
0 2	0	lo			
fe80::42:44ff:fe54:2a00/128				::	Un
0 2	0	eth0			
fe80::42:44ff:fe54:2a01/128				::	Un
0 3	0	eth1			

- Проверка на узле PC3-arromichina настройки DNS и получение адреса по DHCPv6.

```
/ # cat /etc/resolv.conf  
/ # dhclient -6 -v eth0
```

Настройка DHCP в случае IPv6

- Проверка на узле PC3-arromichina настройки сети, пинг маршрутизатора и проверка настройки.

```
PC3-arromichina - PuTTY
eth1    Link encap:Ethernet  HWaddr 02:42:44:54:2A:01
        inet6 addr: fe80::42:44ff:fe54:2a01/64 Scope:Link
        UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
        RX packets:0 errors:0 dropped:0 overruns:0 frame:0
        TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
        collisions:0 txqueuelen:1000
        RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)

lo      Link encap:Local Loopback
        inet addr:127.0.0.1  Mask:255.0.0.0
        inet6 addr: ::1/128 Scope:Host
        UP LOOPBACK RUNNING  MTU:65536  Metric:1
        RX packets:0 errors:0 dropped:0 overruns:0 frame:0
        TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
        collisions:0 txqueuelen:1000
        RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)

/ # route -n -A inet6
Kernel IPv6 routing table
Destination                                Next Hop
  Flags Metric Ref  Use Iface
fe80::/64
U        256   1    0 eth0
fe80::/64
U        256   1    0 eth1
::/0
UGDA    1024   1    0 eth0
fe80::eae:92ff:fe72:2
::1/128
Un       0     2    0 lo
fe80::42:44ff:fe54:2a00/128
Un       0     2    0 eth0
fe80::42:44ff:fe54:2a01/128
Un       0     3    0 eth1
ff00::/8
U        256   3    0 eth0
ff00::/8
U        256   1    0 eth1
::/0
In      -1     1    0 lo
```


Настройка DHCP в случае IPv6

- Просмотр на маршрутизаторе статистики DHCP-сервера и выданных адресов.

```
arromichina@arromichina-gw-01# run show dhcpv6 server leases
IPv6 address      State      Last communication    Lease expiration    Remaining    Type    Pool
IAID_DUID
-----
[edit]
arromichina@arromichina-gw-01#
```

Настройка DHCP в случае IPv6

- Проанализируем захваченные анализатором трафика пакеты

л Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

Примените фильтр отображения ... «Ctrl-/»

Time	Source	Destination	Protocol	Length	Info
1 0.000000	::	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
2 0.135850	::	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
3 0.719734	::	ff02::1:ff54:2a00	ICMPv6	86	Neighbor Solicitation for fe80::42:44ff:fe54:2a00
4 1.743988	fe80::42:44ff:fe54::...	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
5 1.744512	fe80::42:44ff:fe54::...	ff02::2	ICMPv6	70	Router Solicitation from 02:42:44:54:2a:00
6 1.952229	fe80::42:44ff:fe54::...	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
7 5.872073	fe80::42:44ff:fe54::...	ff02::2	ICMPv6	70	Router Solicitation from 02:42:44:54:2a:00
8 14.319889	fe80::42:44ff:fe54::...	ff02::2	ICMPv6	70	Router Solicitation from 02:42:44:54:2a:00
9 30.191962	fe80::42:44ff:fe54::...	ff02::2	ICMPv6	70	Router Solicitation from 02:42:44:54:2a:00
10 62.703874	fe80::42:44ff:fe54::...	ff02::2	ICMPv6	70	Router Solicitation from 02:42:44:54:2a:00
11 116.096424	fe80::eae:92ff:fe72::...	ff02::16	ICMPv6	150	Multicast Listener Report Message v2
12 116.103119	fe80::eae:92ff:fe72::...	ff02::16	ICMPv6	170	Multicast Listener Report Message v2
13 116.301917	::	ff02::1:ff00:1	ICMPv6	86	Neighbor Solicitation for 2001::1
14 116.925887	fe80::eae:92ff:fe72::...	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
15 124.144122	fe80::42:44ff:fe54::...	ff02::2	ICMPv6	70	Router Solicitation from 02:42:44:54:2a:00
16 251.120490	fe80::42:44ff:fe54::...	ff02::2	ICMPv6	70	Router Solicitation from 02:42:44:54:2a:00
17 509.167184	fe80::42:44ff:fe54::...	ff02::2	ICMPv6	70	Router Solicitation from 02:42:44:54:2a:00
18 1033.452747	fe80::42:44ff:fe54::...	ff02::2	ICMPv6	70	Router Solicitation from 02:42:44:54:2a:00
19 1086.003326	fe80::eae:92ff:fe72::...	ff02::16	ICMPv6	110	Multicast Listener Report Message v2
20 1086.782336	fe80::eae:92ff:fe72::...	ff02::16	ICMPv6	110	Multicast Listener Report Message v2
21 1088.613227	fe80::eae:92ff:fe72::...	ff02::1	ICMPv6	86	Router Advertisement from 0c:ae:92:72:00:02
22 1104.651153	fe80::eae:92ff:fe72::...	ff02::1	ICMPv6	86	Router Advertisement from 0c:ae:92:72:00:02
23 1120.670247	fe80::eae:92ff:fe72::...	ff02::1	ICMPv6	86	Router Advertisement from 0c:ae:92:72:00:02
24 1255.929548	fe80::42:44ff:fe54::...	2001::1	ICMPv6	118	Echo (ping) request id=0x3201, seq=0, hop limit=64 (reply in 25)
25 1255.930569	2001::1	fe80::42:44ff:fe54::...	ICMPv6	118	Echo (ping) reply id=0x3201, seq=0, hop limit=64 (request in 24)
26 1256.930004	fe80::42:44ff:fe54::...	2001::1	ICMPv6	118	Echo (ping) request id=0x3201, seq=1, hop limit=64 (reply in 27)

Frame 12: 170 bytes on wire (1360 bits), 170 bytes captured (1360 bits) on interface -, id 0

Ethernet II, Src: 0c:ae:92:72:00:02 (0c:ae:92:72:00:02), Dst: IPv6mcast_16 (33:33:00:00:00:16)

Internet Protocol Version 6, Src: fe80::eae:92ff:fe72:2, Dst: ff02::16

Internet Control Message Protocol v6

0000 33 33 00 00 00 16 0c ae 92 72 00 02 86 dd 60 00

0010 00 00 00 74 00 01 fe 80 00 00 00 00 00 00 0e ae

0020 92 ff fe 72 00 02 ff 02 00 00 00 00 00 00 00 00

0030 00 00 00 00 00 16 3a 00 05 02 00 00 01 00 8f 00

0040 c6 06 00 00 00 05 04 00 00 00 ff 02 00 00 00 00

0050 00 00 00 00 00 01 ff 00 00 01 04 00 00 00 ff 02

Вывод

- В ходе выполнения лабораторной работы мы получили навыки настройки службы DHCP на сетевом оборудовании для распределения адресов IPv4 и IPv6.